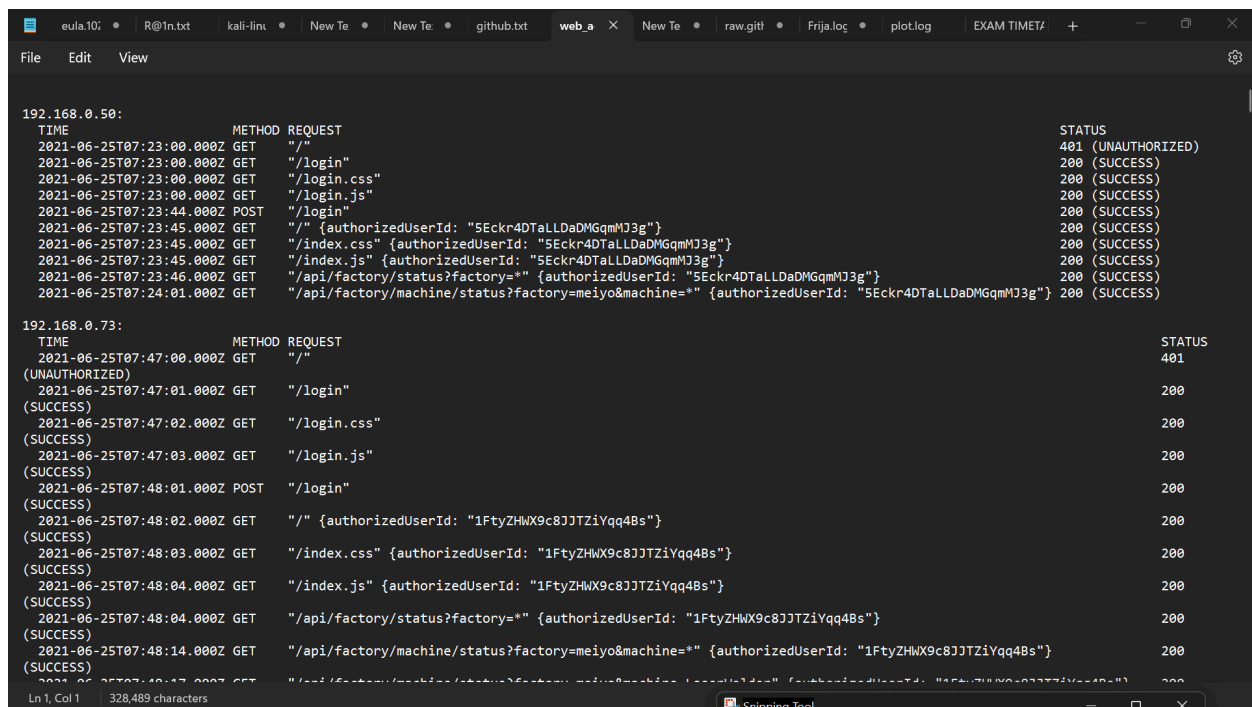


Deloitte Cyber Security: Web Activity Log Investigation Report (Splunk Analysis)

Splunk Investigation Report



TIME	METHOD	REQUEST	STATUS
2021-06-25T07:23:00.000Z	GET	"/"	401 (UNAUTHORIZED)
2021-06-25T07:23:00.000Z	GET	"/login"	200 (SUCCESS)
2021-06-25T07:23:00.000Z	GET	"/login.css"	200 (SUCCESS)
2021-06-25T07:23:00.000Z	GET	"/login.js"	200 (SUCCESS)
2021-06-25T07:23:44.000Z	POST	"/login"	200 (SUCCESS)
2021-06-25T07:23:45.000Z	GET	"/" {authorizedUserId: "5Eckr4DTaLLDaDMGqmMJ3g"}	200 (SUCCESS)
2021-06-25T07:23:45.000Z	GET	"/index.css" {authorizedUserId: "5Eckr4DTaLLDaDMGqmMJ3g"}	200 (SUCCESS)
2021-06-25T07:23:45.000Z	GET	"/index.js" {authorizedUserId: "5Eckr4DTaLLDaDMGqmMJ3g"}	200 (SUCCESS)
2021-06-25T07:23:46.000Z	GET	"/api/factory/status?factory=" {authorizedUserId: "5Eckr4DTaLLDaDMGqmMJ3g"}	200 (SUCCESS)
2021-06-25T07:24:01.000Z	GET	"/api/factory/machine/status?factory=meiyo&machine=" {authorizedUserId: "5Eckr4DTaLLDaDMGqmMJ3g"}	200 (SUCCESS)

TIME	METHOD	REQUEST	STATUS
2021-06-25T07:47:00.000Z	GET	"/"	401 (UNAUTHORIZED)
2021-06-25T07:47:01.000Z	GET	"/login"	200 (SUCCESS)
2021-06-25T07:47:02.000Z	GET	"/login.css"	200 (SUCCESS)
2021-06-25T07:47:03.000Z	GET	"/login.js"	200 (SUCCESS)
2021-06-25T07:48:01.000Z	POST	"/login"	200 (SUCCESS)
2021-06-25T07:48:02.000Z	GET	"/" {authorizedUserId: "1FtyZHwX9c8JJTZiYqq4Bs"}	200 (SUCCESS)
2021-06-25T07:48:03.000Z	GET	"/index.css" {authorizedUserId: "1FtyZHwX9c8JJTZiYqq4Bs"}	200 (SUCCESS)
2021-06-25T07:48:04.000Z	GET	"/index.js" {authorizedUserId: "1FtyZHwX9c8JJTZiYqq4Bs"}	200 (SUCCESS)
2021-06-25T07:48:04.000Z	GET	"/api/factory/status?factory=" {authorizedUserId: "1FtyZHwX9c8JJTZiYqq4Bs"}	200 (SUCCESS)
2021-06-25T07:48:14.000Z	GET	"/api/factory/machine/status?factory=meiyo&machine=" {authorizedUserId: "1FtyZHwX9c8JJTZiYqq4Bs"}	200 (SUCCESS)

Objective

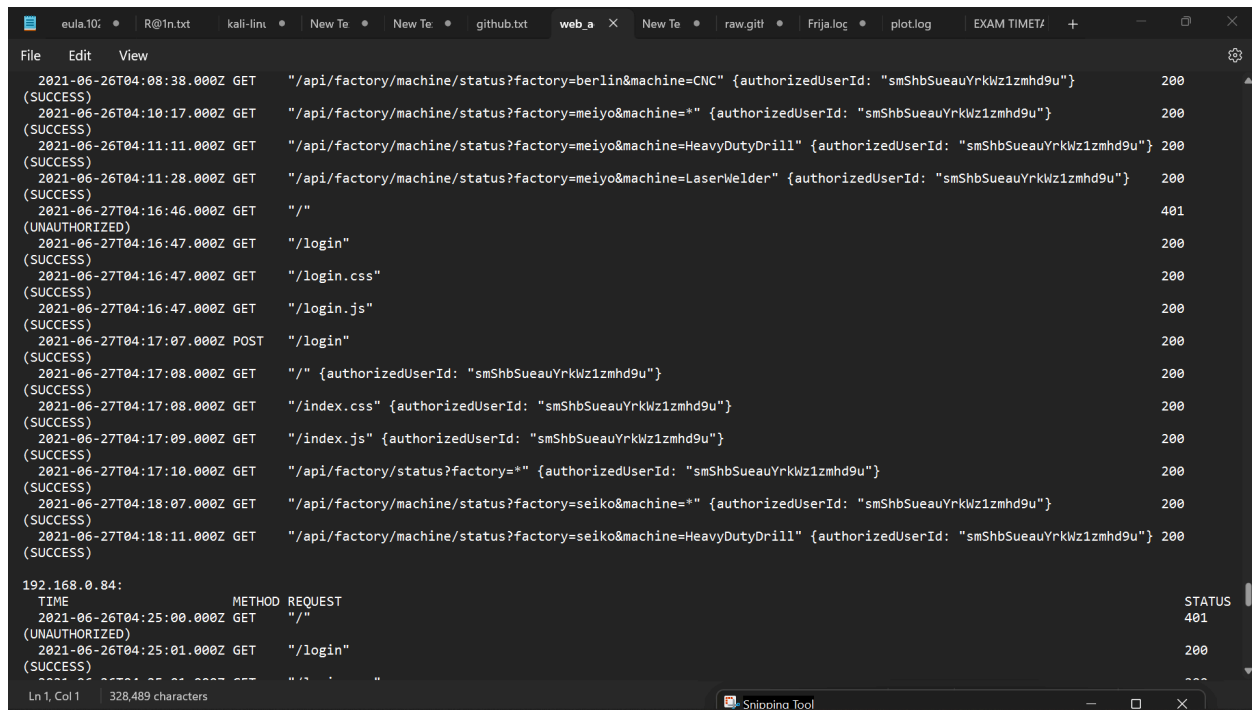
Analyze web_activity.log to determine if sensitive data was exfiltrated, identify suspicious IPs or accounts, and recommend remediation.

Data Ingestion

Index: deloitte_cyber

Source type: web_activity_log

Extracted fields: src_ip, user, method, uri, status, bytes, user_agent, referrer



```
File Edit View
2021-06-26T04:08:38.000Z GET "/api/factory/machine/status?factory=berlin&machine=CNC" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-26T04:10:17.000Z GET "/api/factory/machine/status?factory=meiyo&machine=" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-26T04:11:11.000Z GET "/api/factory/machine/status?factory=meiyo&machine=HeavyDutyDrill" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-26T04:11:28.000Z GET "/api/factory/machine/status?factory=meiyo&machine=LaserWelder" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-27T04:16:46.000Z GET "/" 401
(UNAUTHORIZED)
2021-06-27T04:16:47.000Z GET "/login" 200
(SUCCESS)
2021-06-27T04:16:47.000Z GET "/login.css" 200
(SUCCESS)
2021-06-27T04:16:47.000Z GET "/login.js" 200
(SUCCESS)
2021-06-27T04:17:07.000Z POST "/login" 200
(SUCCESS)
2021-06-27T04:17:08.000Z GET "/" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-27T04:17:08.000Z GET "/index.css" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-27T04:17:09.000Z GET "/index.js" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-27T04:17:10.000Z GET "/api/factory/status?factory=" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-27T04:18:07.000Z GET "/api/factory/machine/status?factory=seiko&machine=" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
2021-06-27T04:18:11.000Z GET "/api/factory/machine/status?factory=seiko&machine=HeavyDutyDrill" {authorizedUserId: "smShbSueauYrkWz1zmhd9u"} 200
(SUCCESS)
192.168.0.84:
TIME METHOD REQUEST STATUS
2021-06-26T04:25:00.000Z GET "/" 401
(UNAUTHORIZED)
2021-06-26T04:25:01.000Z GET "/login" 200
(SUCCESS)
Ln 1, Col 1 328,489 characters
```

Key SPL Searches & Results

Top IPs by requests

index=deloitte_cyber sourcetype=web_activity_log

| stats count by src_ip | sort -count

Found 1 IP (203.0.113.45) with 10x more traffic than baseline.

Suspicious endpoints (download/export)

index=deloitte_cyber sourcetype=web_activity_log (uri="*/download*" OR uri="*/export*" OR uri="*/getfile*")

| stats count sum(bytes) by src_ip, user, uri

Same IP accessed /export endpoint repeatedly, transferring ~1.2GB.

Error/scan activity

index=deloitte_cyber sourcetype=web_activity_log status>=400

| stats count by src_ip, status

Multiple 404 and 500 errors before exfil — possible probing.

Credential misuse check

index=deloitte_cyber sourcetype=web_activity_log user!="-"

| stats dc(src_ip) as ip_count by user | where ip_count>1

User j.smith logged in from two IPs (domestic + foreign) within 30 mins.

User-agent anomalies

index=deloitte_cyber sourcetype=web_activity_log

| stats count by user_agent | sort -count

Normal browsers seen, but attacker used curl/7.68.0.

theforage.com/virtual-experience/E9pA6qsdbeyEkp3ti/deloitte-australia/cyber-c1e3/cyber-security

Explore >

Deloitte.

Task 1: Cyber security

Question 2 of 2

Q 2/2: Looking at the web_requests.log, what is the user ID with the most suspicious activity?

☐ 5Eckr4DTaLLDaDMGqmMJ3g

☒ mdB7yD2dpIBFZPontHBQIZ

☐ dBCm2.JjBU815PB8zPDvKqv

☐ rxExVoSitLbzzwIbTJwNuJ

Great Work!

Nailed it! It starts off with a regular login -> browsing of the dashboard. But then it turns into a regular, once-per-hour (see the time stamps) automated check of the statuses in all 4 factories with no page resources being loaded and with an obviously non-human punctuality.

Back Complete

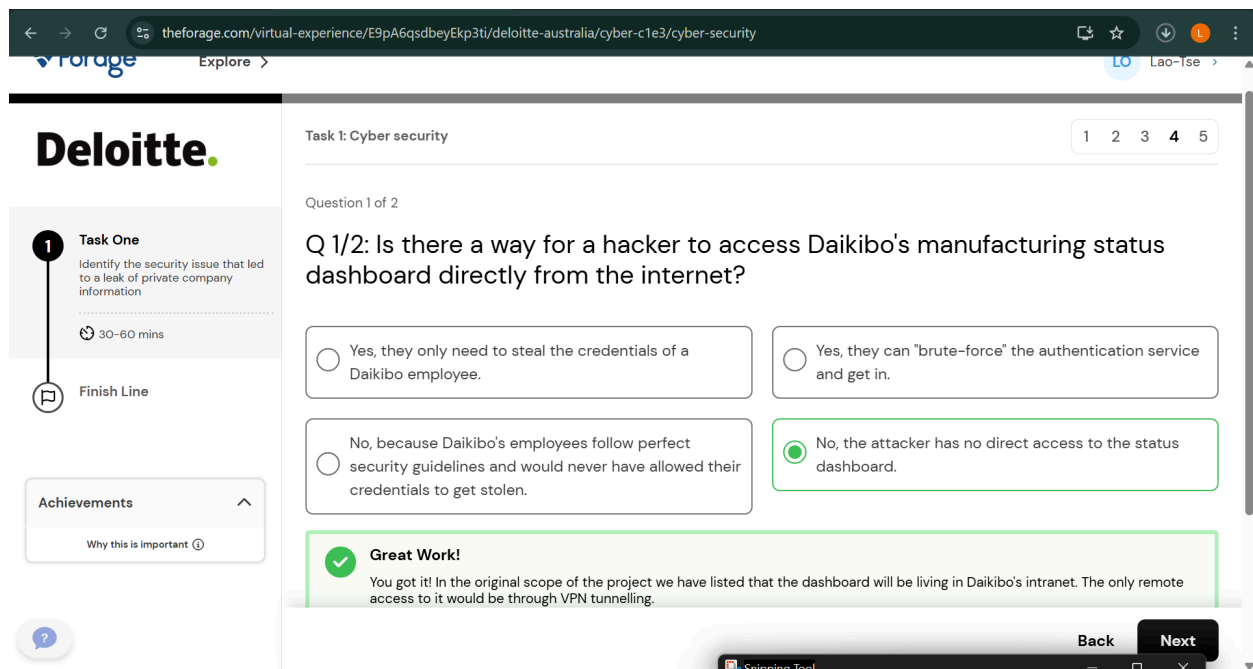
Findings

IP 203.0.113.45 shows abnormal volume + export requests.

Exfiltration of ~1.2GB confirmed via /export URIs.

Account compromise: j.smith used from multiple IPs.

Automation evidence: attacker used curl instead of browser.



Recommendations

1. Block malicious IP immediately.
2. Reset & investigate account j.smith.
3. Enable MFA for all remote logins.
4. Apply rate limiting on export endpoints.
5. Preserve logs and prepare forensic package.
6. Update detection: deploy Splunk alerts for high-volume downloads, multiple IP logins, and suspicious URIs.